

Start here: viva readiness checklist

Thirty-second description

I compared a same-backbone AE and VAE on CelebA, showed why only the VAE can sample from a normal prior, trained an MNIST denoising AE, deliberately produced vanishing gradients and mode collapse in GANs, then trained DCGAN and WGAN-GP alternatives. I evaluated the saved runs with a six-digit classifier, reverse KL, my own IS and classifier-feature FID. The key evidence is the C1 gradient dying at iteration 448, complete C3 collapse to digit 3, and D2 reaching FID 17.117 while covering all six modes.

Facts to know without looking

Item	Exact fact
Roll / R / seed	20L-11327 / 27 / 1327
Assigned values	held-out 7; digits [0, 2, 3, 6, 7, 9]; Blond_Hair plus Smiling
Main run	20L-11327_main_v1, CUDA
CelebA split	27,000 train / 3,000 validation
AE/VAE	latent 64, Adam 1e-3, batch 64, 10 epochs
AE validation MSE	0.0066048581
VAE validation MSE	0.0083013152; gap 0.0016964571
Denoising PSNR	13.3204 -> 19.4447 dB; gain about 6.124 dB
Classifier	99.0194% on the six assigned digits
C1 reference	iter 100 first-layer norm 7.0218186e-05
C1 threshold/death	1% = 7.0218186e-07; first crossing iter 448, norm 0
C2 at iter 448	first-layer norm 3.83656; full norm 11.29571
C3	counts [0, 0, 5000, 0, 0, 0]; digit 3; 1 mode; reverse KL ln(6)=1.791759
D1	6 modes; reverse KL .052779; IS 4.4772 +/- .1080; FID 39.700
D2	6 modes; reverse KL .019803; IS 4.6190 +/- .0898; FID 17.117
Real evaluation	IS 5.7720 +/- .0262; FID 1.39865
C1 / C3 FID	941.881 / 821.241
FID floor	1.200874
Noise FID	sigma 0/.1/.25/.5/1 -> 0/14.144/100.552/326.750/543.036

Checklist

- [] Derive the VAE KL term and reparameterization without notes.
- [] Explain the exact loss scaling used in the code and why beta is still 1.
- [] Explain the A3 control: AE own codes work, normal prior codes do not.
- [] Derive PSNR from MSE for values in [0, 1].
- [] Derive $d \log(1-D)/da = -D$ and $d[-\log D]/da = -(1-D)$.
- [] Open evidence/C1.csv, point to rows 100, 447, and 448, and calculate the threshold.
- [] Explain why the 447-to-448 transition is abrupt rather than a smooth decay.
- [] Distinguish g_grad_norm (first generator layer) from g_full_grad_norm.
- [] Recompute C3 reverse KL from the six class counts.
- [] State that actual digit 3 is remapped class index 2.
- [] Explain the 1% mode-coverage rule: at least 50 of 5,000 samples.
- [] Trace D1 architecture and explain [0, 1] <-> [-1, 1] scaling.
- [] Write the WGAN-GP critic loss, generator loss, interpolation, and penalty.
- [] Explain why WGAN uses critic scores, not probabilities, and no critic BatchNorm.
- [] Derive IS and explain its two desired properties.
- [] Derive FID and identify the exact feature/reference sets.
- [] Explain the non-zero FID floor and the sigma=0 near-zero result.
- [] Explain the singular-covariance warning for collapsed samples.
- [] Name at least three limitations without weakening valid conclusions.

- [] Explain the handout/starter inconsistencies without claiming extra work.
- [] State the disclosed AI assistance honestly and demonstrate ownership live.

The highest-risk examiner probes

1. Why was iteration 448 selected, and is it based on the full gradient?
2. What exactly changed between C1 and C2?
3. Why does the VAE loss divide KL by the pixel count?
4. Why can a collapsed covariance make $\sqrt{\text{tm}}$ numerically awkward?
5. Are D2's D_x and D_{G_z} probabilities?
6. Did every run literally reset to seed 1327?
7. Why are A4, B2, and Part F absent?
8. Which parts of the code can you write unaided during the viva?

Study order

First memorize the facts table. Then do the five derivations in 02_DERIVATIONS_AND_NUMERICALS.md. Next perform every live drill in 03_CODE_AND_EVIDENCE_WALKTHROUGH.md. Finish with both mock rounds without reading answers.

TA review and risk register

Overall verdict before this package

The strict TA assessment judged the submission technically strong and approximately 40.5/41 on the visible marks. Report, notebook, logs, figures, metrics, and assigned values are mutually consistent. The viva risk is ownership and precision, not missing core experiments.

Checks performed

- Read the complete handout and four-page report.
- Inspected the 36-cell executed notebook and matching Python source.
- Checked row counts and columns for every AE/VAE, denoising, classifier, GAN, IS, FID, and preflight log.
- Verified exact C1 rows 100 and 448 and the C2 matched row.
- Verified C3/D1/D2 counts, coverage, and reverse KL values.
- Verified all values in `metrics.json`, `E_summary.csv`, and the report.
- Checked the WGAN-GP, IS, FID, and validation code paths.
- Reviewed reproducibility metadata, fixed indices, fixed noise, and LLM declaration.

Strengths

- Numerical claims are backed by raw saved evidence rather than generic explanations.
- Negative results are preserved and explained honestly.
- C1/C2 is a controlled one-line loss comparison.
- Both required and full-generator gradient norms are logged.
- The classifier has an accuracy gate and confusion matrix.
- IS split scores and FID diagnostics are saved.
- Preflight overfits and smoke tests demonstrate a natural debugging process.
- Report fits the four-page limit and directly uses the measured run.

High-risk viva issues

1. Row 447-to-448 is abrupt; never describe a smooth monotone gradient death.
2. The death threshold is based on the first generator layer, not the full norm.
3. VAE KL scaling must be explained as fixed objective scaling, not beta reduction.
4. WGAN critic columns are scores, not probabilities.
5. A singular covariance warning is expected under collapse but still deserves a numerical caveat.
6. Seed offsets are reproducible but not literal compliance with resetting 1327 each run.
7. D2 is not a one-factor ablation of gradient penalty.
8. A2 is entangled and C3 lacks a quantitative temporal stopping criterion.
9. AI assistance is substantial and disclosed, so code ownership will be tested.

Handout/starter traps

- A1-A3 sum to 8 although Part A says 12 marks.
- A4, B2, and Part F are referenced but absent.
- The footer says Assignment 2 while the title says Assignment 1.
- The requested filename is `A2_<roll>.ipynb`.
- The handout requires D BatchNorm for D1, while the starter base discriminator omitted it; the submission added it.
- The starter-compatible norm is first-layer, while the prose can sound like full-generator norm; the submission logged both.
- A built-in spectral-normalization option would violate "implement it"; the submission instead implements WGAN-GP manually.

Derivations and numerical checks

1. AE and VAE objectives

For an AE, $z = \text{encoder}(x)$ and $x_{\text{hat}} = \text{decoder}(z)$. The submitted reconstruction quantity is mean squared pixel error:

$$\text{MSE} = (1 / (N * C * H * W)) * \sum((x_{\text{hat}} - x)^2).$$

For a diagonal Gaussian posterior, the encoder returns μ and $\log\text{var} = \log(\sigma^2)$. The reparameterized sample is:

$$\epsilon \sim N(0, I), z = \mu + \exp(0.5 * \log\text{var}) * \epsilon.$$

Randomness is isolated in ϵ , while z remains differentiable with respect to μ and $\log\text{var}$.

Against $p(z) = N(0, I)$, the KL for one example is:

$$\text{KL}(q||p) = 0.5 * \sum(\exp(\log\text{var}) + \mu^2 - 1 - \log\text{var}).$$

The code averages KL across the batch and uses:

$$\text{loss} = \text{pixel_mean_MSE} + \text{KL_per_image} / (3 * 64 * 64).$$

Multiplying by $3 * 64 * 64$ gives per-image SSE + KL, so this is $\beta=1$ ELBO scaling divided by one fixed positive constant. It changes the displayed scale, not the optimizer's minimizer.

Measured comparison: AE validation MSE 0.0066048581; VAE 0.0083013152; VAE reconstruction gap 0.0016964571, about 25.7% relative to the AE value. The KL regularizes the latent distribution, trading some reconstruction sharpness for meaningful prior sampling.

2. Why the A3 control matters

The AE decoder succeeds on codes produced by its own encoder but fails on the same $N(0, I)$ draws that produce face-like VAE samples. Therefore the AE decoder is not simply broken. The AE aggregate posterior was never constrained to match the normal prior, so arbitrary normal samples land off the learned code manifold.

VAE interpolation uses posterior means, not random posterior samples:

$$z_t = (1-t) * \mu_1 + t * \mu_2, \text{ for } t=0, .1, \dots, 1.$$

Using means removes sampling noise. The strips remain plausible, but smile, hair, and identity change together, so the direction is smooth but not disentangled or causal.

3. PSNR

For data in $[0, 1]$:

$$\text{PSNR} = 10 * \log_{10}(1/\text{MSE}).$$

The noisy baseline was 13.3204 dB; the denoised output was 19.4447 dB, a gain of 6.1242 dB. A denoiser is conditional: the corrupted input anchors the output near the data manifold. It does not need an unconditional sampling prior.

4. Logistic GAN losses and gradient death

Let a be the discriminator fake-sample logit and $D = \text{sigmoid}(a)$.

Saturating/minimax generator loss:

$$L_{\text{sat}} = \log(1-D).$$

Since $dD/da = D(1-D)$, then:

$$dL_{\text{sat}}/da = -D.$$

When a strong discriminator gives fake samples $D \approx 0$, the generator's signal vanishes.

Non-saturating loss:

$$L_{\text{ns}} = -\log(D) \text{ and } dL_{\text{ns}}/da = -(1-D).$$

When $D \approx 0$, this derivative is approximately -1. Both objectives have the same ideal generator fixed point but very different optimization fields away from equilibrium.

The stable code forms are $-\text{softplus}(a)$ for the saturating objective and $\text{softplus}(-a)$ for the non-saturating objective.

Run-specific threshold

- Iteration 100 first-layer gradient norm: $7.021818601e-05$.
- One percent: $7.021818601e-07$.
- First later row below the threshold: iteration 448.

- At 448: first-layer norm 0, full-generator norm 0, and $D(G(z))=0$.
- C2 at the same iteration: first-layer norm 3.836562, full norm 11.295707.

The trace is not a perfectly smooth decay. Row 447 has a sharp numerical excursion before row 448 reaches zero. Describe the measured first crossing, not an invented smooth story.

5. Gradient norm definitions

The starter hook measures the first parameter tensor in $G.fc$, while the handout wording refers more generally to the generator-gradient L2 norm:

```
g_grad_norm = ||grad(first G.fc parameter)||_2.
```

The submission also records:

```
g_full_grad_norm = sqrt(sum_p ||grad(p)||_2^2).
```

The reported death threshold uses the starter's first-layer norm. The submission additionally logs the full-generator norm as supporting evidence; it also equals zero at iteration 448.

6. Mode coverage and reverse KL

With 5,000 samples, a mode is covered if its class proportion is at least 1%, meaning at least 50 samples.

For target $q_i=1/6$ and generated proportions p_i :

```
D_KL(p_gen || p_data) = sum_{i:p_i>0} p_i*log(p_i/(1/6)).
```

C3 counts are $[0, 0, 5000, 0, 0, 0]$, so $p=[0, 0, 1, 0, 0, 0]$ and:

```
KL = 1*log(1/(1/6)) = log(6) = 1.791759.
```

Terms with $p_i=0$ contribute zero by the limit $p \log p \rightarrow 0$. Digit 3 is the actual digit and class index 2 in $[0, 2, 3, 6, 7, 9]$.

7. DCGAN

The generator maps $(B, 64)$ to $(B, 128, 7, 7)$, then transposed convolutions to $(B, 64, 14, 14)$ and $(B, 1, 28, 28)$. Tanh produces $[-1, 1]$, matching real images transformed by $2 \times x - 1$.

The discriminator maps $(B, 1, 28, 28)$ to 14×14 and 7×7 feature maps, then one logit. D1 uses strided convolutions, G BatchNorm, D BatchNorm after its second convolution, ReLU in G, LeakyReLU(0.2) in D, Adam $2e-4$ with $\beta_1 = .5$, non-saturating loss, and 1:1 updates.

Fixed z at four checkpoints controls sample identity, so the progression shows learning rather than cherry-picking different random samples.

8. WGAN-GP

The critic has no sigmoid. Its outputs are unrestricted scores, so D2's D_x and D_{G_z} log columns are not probabilities.

Critic objective minimized by the code:

```
L_C = mean(C(fake)) - mean(C(real)) + lambda*GP.
```

Generator objective:

```
L_G = -mean(C(fake)).
```

For $\alpha \sim \text{Uniform}(0, 1)$:

```
x_hat = alpha*x_real + (1-alpha)*x_fake.
```

```
GP = mean((||grad_x_hat C(x_hat)||_2 - 1)^2).
```

The run uses five critic steps, $\lambda=10$, Adam $1e-4$, betas $(0, .9)$, and no critic BatchNorm. BatchNorm couples examples, making the per-example input-gradient constraint ambiguous.

D2 ties D1 at six covered modes and improves balance, IS, and FID. Say it improved the reported distribution metrics, not that it was universally superior under all possible criteria.

9. Inception Score

For each generated sample the six-digit classifier supplies $p(y|x)$. Within each split:

```
p(y) = mean_x p(y|x).
```

```
IS = exp(mean_x KL(p(y|x) || p(y))).
```

High IS requires confident conditional predictions and a diverse marginal. The implementation splits 5,000 samples into 10 groups of 500 and reports mean and population standard deviation over split scores.

C3 has IS 1.0006: every image is assigned the same class, so $p(y|x)$ is nearly equal to $p(y)$ and KL is near zero. IS never compares generated images with real images, so it can miss realism failures not exposed by its classifier.

10. Classifier-feature FID

The classifier's 128-dimensional penultimate features give real and generated matrices of shape (5000, 128). For means and covariances:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}).$$

The generated sets are compared against a seeded 5,000-image real training reference. The real row in `E_summary.csv` is a separate real test subset evaluated against that training reference, so its FID is not zero.

$\Sigma_r \Sigma_g$ is generally nonsymmetric. `scipy.linalg.sqrtm` may return tiny imaginary components from floating-point error; the code accepts only small components, takes the real part, retries with diagonal epsilon for nonfinite or large-imaginary results, and clamps tiny negative final distances to zero.

Collapsed samples make feature covariance low-rank, which can trigger a singular-matrix warning. The values remained finite and the saved diagnostic showed zero imaginary component and no epsilon retry. A warning is not proof of correctness; diagonal regularization or a symmetric PSD formulation would be a stronger robustness check.

FID models only the first two moments of chosen features. It can miss higher-order differences and artifacts the classifier features ignore.

11. FID sanity checks

- Floor: two disjoint real test halves give 1.200874, measuring finite-sample differences in empirical moments.
- Monotonicity: the same clean images plus clipped Gaussian noise give FID approximately 0, 14.144, 100.552, 326.750, 543.036 as sigma increases.
- Sigma 0 compares identical images/features, so only numerical roundoff remains: 4.84e-09.

12. Complexity and reproducibility

Convolutional training cost is dominated by forward/backward convolutions and scales with batch size, spatial dimensions, kernel area, and channel products. WGAN-GP costs substantially more because it performs five critic updates per generator update and a second-order autograd graph for the input-gradient penalty.

The base seed is 1327, while documented offsets create deterministic substreams for data/noise/evaluation. This avoids accidental reuse and makes artifacts reproducible, but it is not literal compliance with "seed 1327 at the top of every run." State this limitation plainly.

Code and evidence walkthrough

Executed notebook map

Notebook cell	Purpose	Must explain
2	Setup and reproducibility	roll parsing, seed, device, paths
4	Evidence helpers	CSV writing, finite checks, gradient norms, preflight gates
6	Data	CelebA cache/split, digit restriction and remapping
8	AE/VAE	common backbone, reparameterization, KL scaling, training
10	A2	endpoint selection and mean-code interpolation
12	A3	shared prior codes and AE own-code control
14	B1	corruption, denoiser, PSNR
16	Classifier	six-class mapping, 128-D features, accuracy gate
18	GAN core	G/D shapes, stable losses, logging, fixed z
20	C1	saturating loss and gradient death
22	C2	one-line loss change and matched comparison
24	Mode helpers	sample collection, histogram, coverage, reverse KL
26	C3	forced-collapse configuration
28	D1	DCGAN configuration and fixed progression
30	D2	manual WGAN-GP implementation
32	E	manual IS/FID and sanity checks
34	Validator	required files, columns, finite values

Matching Python source map

Lines	Topic
45-118	roll values, device, and seed setup
138-225	evidence helpers and checkpoint metadata
270-388	data wrappers, CelebA cache, fixed split
390-490	AE/VAE architecture, KL, reconstruction epoch
520-674	A1 training, A2 interpolation, A3 sampling
681-788	denoising AE and PSNR
799-902	six-digit classifier
917-974	G28, D28, and per-iteration logger
1005-1073	logistic GAN loop and death detection
1196-1248	classifier sampling and mode measurements
1254-1280	C3 and D1 configurations
1286-1358	WGAN-GP penalty and training loop
1364-1404	IS and FID implementations
1415-1493	evaluation sets, floor, and noise sanity check
1499-1556	package validation and final checks

Evidence map

Claim	Open this file	What to point at
Assigned values/configs	run_config.json	R, seed, digits, attribute, every run setting
AE/VAE values	evidence/A1_ae.csv, evidence/A1_vae.csv	final epoch train/validation rows
A2 endpoints	evidence/A2_endpoints.json	saved indices and 11 t values
Denoising	evidence/B1_denoise.csv, metrics.json	final val MSE and both PSNR values
Classifier reliability	evidence/classifier.csv, evidence/classifier_confusion.csv	99.019% and six-class errors
C1 death	evidence/key_rows.csv, evidence/C1.csv	rows 100, 447, 448
C1/C2 mechanism	figures/C2_overlay.png, evidence/C2.csv	matched iteration 448
C3 collapse	evidence/C3_counts.csv, figures/C3_hist.png	digit 3 count 5000
D1 balance	evidence/D1_counts.csv, figures/D1_progress.png	all modes, fixed-z progression

Claim	Open this file	What to point at
D2 improvement	evidence/D2_counts.csv, evidence/final_model_comparison.csv	reverse KL, IS, FID
IS split arithmetic	evidence/E1_split_scores.csv	10 scores for each of five sets
FID sets	evidence/E_summary.csv	real/C1/C3/D1/D2 rows
FID monotonicity	evidence/E3_noise_fid.csv	five increasing rows
Process/authenticity	evidence/experiment-diary.md, preflight_*.csv	natural checks and observations

Live drill 1: prove iteration 448

1. Open evidence/C1.csv.
2. At iteration 100 point to $g_grad_norm=7.0218186e-05$.
3. Calculate one percent: $7.0218186e-07$.
4. Search later rows and show 448 is the first value below the threshold.
5. Point to first-layer and full norms both being zero at 448.
6. Inspect row 447 so you do not falsely describe a smooth decline.
7. Open evidence/C2.csv at 448 and contrast $3.836562 / 11.295707$.

Live drill 2: recompute collapse metrics

1. Open evidence/C3_counts.csv.
2. Map class index 2 to actual digit 3.
3. Coverage threshold is 50 samples; only one row passes.
4. Compute $\log(1/(1/6)) = \log(6) = 1.791759$.
5. Explain why zero-mass terms contribute zero.

Live drill 3: trace one IS split

1. Generated classifier probabilities have shape $(5000, 6)$.
2. Each of 10 chunks has shape $(500, 6)$.
3. Its marginal has shape $(1, 6)$.
4. Calculate per-sample KL, average, exponentiate.
5. Locate the corresponding saved score in evidence/E1_split_scores.csv.

Live drill 4: identify FID populations

1. The classifier produces $(5000, 128)$ features.
2. The common reference is a seeded real training subset.
3. The real row is a real test subset compared with training.
4. The floor is separate: two disjoint halves of the restricted real test set.
5. The monotonicity test compares clean images with clipped noisy versions of the same images.

Live drill 5: trace WGAN-GP line by line

In A1_working_source.py, explain lines 1286-1298: interpolation, requires_grad, critic scores, autograd.grad, create_graph, flattening, L2 norm, and squared penalty. Then explain lines 1323-1334: critic objective, five critic updates, parameter freezing, generator objective, gradient logging, and optimizer step.

Commands that need no GPU

```
python practice_quiz.py
python -c "import pandas as pd; d=pd.read_csv('evidence/C1.csv'); print(d[d.iter.isin([100,447,448])])"
python -c "import pandas as pd; print(pd.read_csv('evidence/final_model_comparison.csv'))"
python -c "import json; print(json.load(open('metrics.json'))['E3_fid_floor'])"
```

Natural intermediate work already preserved

- Ten-image AE, VAE, denoiser, and classifier overfit traces.
- A 20-step GAN smoke test with finite losses and gradients.

- Printed tensor shapes/ranges in the executed notebook.
- One CSV row per outer GAN iteration with frequent file flushing.
- Fixed train/validation indices, evaluation indices, and sample seeds.
- Both first-layer and full-generator gradient norms.
- Classifier confusion matrix, raw IS split scores, and FID diagnostics.
- An experiment diary recording planned settings, observations, checks, actions, and evidence.

Do not say

- Do not call D2 critic scores probabilities.
- Do not say C1 gradient decayed smoothly.
- Do not say D2 proves gradient penalty alone caused the improvement.
- Do not claim A2 found a disentangled or causal attribute direction.
- Do not call the FID floor a universal minimum.
- Do not say C3 obtained a respectable IS in this run.
- Do not say every sub-run literally used seed 1327.
- Do not invent the absent A4, B2, or Part F.

Likely viva questions and answer points

Answer directly, show the equation or evidence, then add one limitation. These are answer points, not scripts.

A. Assignment and reproducibility

1. What did you implement?

A same-backbone AE/VAE comparison on CelebA, VAE interpolation and prior sampling, an MNIST denoising AE, deliberate GAN gradient death and mode collapse, a working DCGAN, WGAN-GP, and classifier-based reverse KL, IS, and FID with sanity checks.

2. Derive your assigned values.

$R=27$; seed is last four digits 1327; held-out digit $27 \bmod 10=7$; $(27+3k) \bmod 10$ gives 7, 0, 3, 6, 9, 2, sorted to [0, 2, 3, 6, 7, 9]; $27 \bmod 6=3$, selecting Blond_Hair.

3. What is the main run identity?

20L-11327_main_v1, executed on CUDA with PyTorch 2.11.0+cu128. All report numbers come from this run's saved artifacts.

4. Did every sub-run literally use seed 1327?

The main seed is 1327, but the notebook uses documented deterministic offsets for sub-experiments, samples, and reference sets. That improves isolation and reproducibility but is not literal compliance with the handout's wording; I would state this honestly.

5. Why use fixed noise?

The same latent vectors across checkpoints isolate training progress. Changing z would confound model improvement with different random samples.

6. What did the preflight tests prove?

They checked wiring and capacity, not final quality: AE .08841->.00226, VAE reconstruction .10547->.00240, denoiser .23297->.01328, classifier 1.78433->4.09e-6 on ten images, plus finite GAN gradients over 20 steps.

7. What is odd about the handout numbering?

It references A4, B2, and Parts A-F without defining those tasks; the footer says Assignment 2; and it requests filename A2_<roll>.ipynb. I followed explicit deliverables and did not invent missing work.

8. Why is held-out digit 7 not used?

It appears intended for a missing anomaly task/B2. The supplied B1 is denoising, so inventing an anomaly experiment would not answer a defined question.

B. AE and VAE

9. Trace the CelebA encoder.

$3 \times 64 \times 64 \rightarrow 32 \times 32 \times 32 \rightarrow 64 \times 16 \times 16 \rightarrow 128 \times 8 \times 8 \rightarrow 256 \times 4 \times 4 \rightarrow 4096$. The AE head outputs 64; the VAE head outputs 128 split into 64-dimensional μ and $\log\text{var}$.

10. Trace the decoder.

Latent 64 goes through a linear layer to $256 \times 4 \times 4$, then transposed convolutions produce $128 \times 8 \times 8$, $64 \times 16 \times 16$, $32 \times 32 \times 32$, and $3 \times 64 \times 64$, ending with sigmoid.

11. Derive the VAE KL term.

For one coordinate, $\text{KL}(N(\mu, \sigma^2) || N(0, 1)) = .5 * (\mu^2 + \sigma^2 - 1 - \log(\sigma^2))$. Since $\log\text{var} = \log(\sigma^2)$, code uses $.5 * (\exp(\log\text{var}) + \mu^2 - 1 - \log\text{var})$, sums 64 dimensions, and averages the batch.

12. Why $\exp(0.5 * \log\text{var})$?

$\log\text{var} = \log(\sigma^2)$, so $\sigma = \exp(\log\text{var}/2)$.

13. Why predict log variance?

It is unconstrained on the real line and exponentiation guarantees positive variance. It is also convenient for the analytic KL.

14. Explain reparameterization.

$z = \mu + \sigma * \epsilon$, with $\epsilon \sim N(0, I)$. Randomness is moved into an input independent of encoder parameters, so ordinary backpropagation reaches μ and $\log\text{var}$.

15. Why does validation decode μ ?

It removes Monte Carlo sampling noise and produces deterministic reconstructions. Training samples z , and BatchNorm also changes train/eval behavior, so train and validation reconstruction numbers are not identical inference conditions.

16. How is the VAE loss scaled?

The code uses pixel-mean MSE plus $KL/(3 \times 64 \times 64)$. Multiplying by 12,288 yields per-image SSE+KL, so it is the $\beta=1$ objective divided by a fixed constant, not a smaller β .

17. What were the A1 results?

AE train/validation MSE .00668376/.00660486; VAE .00993852/.00830132. The validation gap is .00169646, about 25.7% above the AE value.

18. Why is the VAE blurrier?

KL trades exact reconstruction capacity for prior-compatible overlapping posteriors, and MSE favors pixel averages. The measured validation gap quantifies this run's tradeoff.

19. Why interpolate means instead of samples?

Means provide stable representative endpoints. Sampled endpoints add epsilon noise and would confound the attribute path.

20. What happened in A2?

Both 11-frame strips stayed face-like. Smile strengthened around/after the middle and hair changed progressively, but identity and hair shape also changed, showing entanglement rather than a pure causal attribute axis.

21. Which A2 endpoints were saved?

Smiling used negative index 22017 and positive 2816. Blond_Hair used negative 2816 and positive 22017, recorded in evidence/A2_endpoints.json.

22. Does a VAE guarantee plausible interpolation or disentanglement?

No. KL encourages a populated continuous latent region, but it is not a proof. The observed identity changes explicitly show the direction is entangled.

23. Why did AE normal-prior samples fail?

The AE decoder only trained on the encoder's aggregate code distribution, which was not constrained to equal $N(0, I)$. Arbitrary normal draws are off-manifold.

24. How do you know the AE decoder itself works?

It decodes codes produced by its own encoder into recognizable faces. A3 shows this control beside the failed normal-prior row.

25. Did the AE output literally "nothing"?

No. The observed outputs were blurred brown face-like averages. I would describe my figure rather than repeat the handout's dramatic wording.

C. Denoising and classifier

26. Define PSNR.

For $[0, 1]$ data, $PSNR = 10 \times \log_{10}(1/MSE)$. The implementation computes per-image values and averages them.

27. What was the denoising result?

Mean PSNR rose from 13.3204 to 19.4447 dB, a 6.1242 dB gain, on one saved seeded 1,000-image test set.

28. Why clip noisy inputs?

MNIST is normalized to $[0, 1]$; clipping $x + 0.3 \times \epsilon$ keeps the corrupted image in the valid input range.

29. Why can an AE denoise but fail at unconditional generation?

Denoising is conditional projection from a nearby corrupted observation toward the learned digit manifold. Unconditional generation requires a known latent prior, which a plain AE lacks.

30. How reliable is the mode classifier?

It reached 99.019% restricted-test accuracy and has a saved confusion matrix. It is strong in-domain, but generated artifacts are out-of-distribution and may receive overconfident labels, so grids remain important.

31. Trace the classifier.

$1 \times 28 \times 28 \rightarrow \text{conv}32/\text{pool} \rightarrow 32 \times 14 \times 14 \rightarrow \text{conv}64/\text{pool} \rightarrow 64 \times 7 \times 7 \rightarrow 3136 \rightarrow 128 \text{ features} \rightarrow 6 \text{ logits}$. The frozen model supplies labels, IS probabilities, and FID features.

D. C1 and C2 gradient death

32. Derive the saturating generator gradient.

For fake logit a , $D = \text{sigmoid}(a)$ and $L_{\text{sat}} = \log(1-D) = -\text{softplus}(a)$. Therefore $dL/da = -D$, which vanishes when D confidently rejects fakes.

33. Derive the non-saturating gradient.

$L_{ns} = -\log D = \text{softplus}(-a)$, so $dL/da = -(1-D)$. When D is approximately 0, the signal remains near -1.

34. Do the two losses have the same objective everywhere?

No. They have the same desired/Nash equilibrium but different scalar objectives and gradient fields away from equilibrium.

35. Why use softplus?

It expresses logistic losses stably in logit space. Explicit $\log(\text{sigmoid}(a))$ or $\log(1-\text{sigmoid}(a))$ can underflow at extreme logits.

36. What exactly changed from C1 to C2?

Only the generator loss branch. Seed, architecture, z_{dim} 64, 3,000 outer iterations, five D steps, one G step, learning rates, and fixed-noise procedure were held constant.

37. Calculate the C1 death threshold.

At iteration 100, first-layer norm was $7.0218186e-05$; one percent is $7.0218186e-07$. Iteration 448 is the first later row below it, with a logged norm of zero.

38. Is the zero gradient mathematically exact?

It is exactly zero in the saved floating-point log, not a claim that the analytic derivative is identically zero. Extreme logits/probabilities can underflow numerically.

39. What is unusual about rows 447 and 448?

The curve has a sharp numerical excursion before the zero at 448; it is not a perfectly smooth decay. The defensible claim is the first threshold crossing defined by the CSV.

40. What gradient norm was used?

g_grad_norm is the L2 norm of the first parameter tensor in G. fc , after backward and before the step. $g_full_grad_norm$ combines all generator parameter gradients. The threshold uses the first-layer hook; both were zero at 448.

41. What was C2's gradient at the matched iteration?

At 448, first-layer norm was 3.836562 and full norm 11.295707, while its samples already resembled digits.

42. Why detach fake images in the D update?

The D update should not accumulate generator gradients. In the G update, D parameters are frozen but autograd still differentiates D's output with respect to its input, so gradients reach G.

43. Why did five D steps and higher D LR hurt C1?

D separated real and fake too quickly, driving fake logits strongly negative. The saturating derivative is proportional to $D(\text{fake})$, so the generator lost its learning signal.

E. Mode collapse and DCGAN

44. What forced C3 to collapse?

The combined recipe: z_{dim} 2, no G BatchNorm, five G updates per D update, $lr_G=2e-3$, and $lr_D=2e-4$. Low z alone is not proven to be the sole cause.

45. What counts as a covered mode?

A predicted class proportion of at least 1%; with 5,000 samples that means at least 50 assignments.

46. Derive C3's reverse KL.

With uniform target and all mass on one class, $KL(p_{gen} || p_{data}) = 1 * \log(1/(1/6)) = \log 6 = 1.791759$. Zero-mass generated classes contribute zero by limit.

47. Why call it reverse KL, and what is its blind spot?

The direction is generated-to-data. Missing real modes with $p_{gen}=0$ contribute nothing directly; forward KL would diverge there. Concentrating mass still produces the maximum $\log 6$ in this six-class case.

48. Which class received all C3 samples?

Actual digit 3, which is remapped class index 2. Counts are $[0, 0, 5000, 0, 0, 0]$ for digits $[0, 2, 3, 6, 7, 9]$.

49. What makes D1 a DCGAN-style model?

Strided convolutions, transposed convolutions, BatchNorm in G and after D's second convolution, ReLU in G, LeakyReLU(.2) in D, tanh output, Adam $2e-4$ with $\beta_1 = .5$, non-saturating loss, and 1:1 updates.

50. Trace the GAN shapes and ranges.

G maps $(B, 64)$ to $128 \times 7 \times 7$, $64 \times 14 \times 14$, then $1 \times 28 \times 28$ in $[-1, 1]$. Real images use 2×-1 ; generated images use $(x+1)/2$ before classifier evaluation. D ends in one logit.

51. What was D1's class balance?

Proportions [.1970, .0740, .2452, .1622, .1470, .1746]; all six modes are covered, but the distribution remains skewed. Reverse KL is .052779.

F. WGAN-GP

52. State the WGAN-GP losses.

Critic minimizes $\text{mean}(C(\text{fake})) - \text{mean}(C(\text{real})) + \lambda * GP$; generator minimizes $-\text{mean}(C(\text{fake}))$. Here λ is 10.

53. Why is it called a critic?

It outputs unrestricted real scores, not probabilities, and has no sigmoid. D2's D_x and D_{G_z} columns are therefore critic scores.

54. Derive the gradient penalty.

Sample α , interpolate $x_{\text{hat}} = \alpha * \text{real} + (1 - \alpha) * \text{fake}$, differentiate $C(x_{\text{hat}})$ with respect to x_{hat} , and penalize $(\| \text{grad } C \|_2 - 1)^2$ averaged over the batch.

55. Why is create_graph=True required?

The critic loss must backpropagate through the calculated input gradient norm, so autograd needs a differentiable graph for that gradient operation.

56. Why enforce 1-Lipschitz behavior?

The Kantorovich-Rubinstein dual form of Wasserstein-1 requires a 1-Lipschitz critic. The penalty softly enforces this near real-fake interpolation paths.

57. Why no critic BatchNorm?

BatchNorm makes one sample's score depend on other batch samples, interfering with a per-sample input-gradient constraint. G still uses BatchNorm.

58. Why five critic steps?

They keep the critic closer to its current optimum so its score difference gives G a useful direction. This is not logistic saturation because the WGAN generator gradient has no sigmoid factor.

59. What were D2's results?

Proportions [.1494, .1594, .2286, .1574, .1850, .1202], six modes, reverse KL .019803, IS 4.61896 ± 0.08984 , and FID 17.11675.

60. Does D2 isolate gradient penalty as the cause?

No. Relative to D1 it also changes objectives, critic-update ratio, optimizer settings, and critic BatchNorm. It demonstrates the WGAN-GP configuration's result, not a one-factor ablation.

G. IS and FID

61. Derive Inception Score.

$IS = \exp(E_x KL(p(y|x) || p(y))) = \exp(H(p(y)) - E_x H(p(y|x)))$. It rewards confident per-sample predictions and a diverse marginal.

62. What is the IS range here?

With six classes, ideal confident balanced predictions approach 6; identical/uninformative predictions give 1. Real data scores 5.772 rather than 6 because of classifier uncertainty and empirical imbalance.

63. How were IS splits computed?

Five thousand samples were divided into 10 disjoint chunks of 500. Each produces a score; NumPy mean and population standard deviation are reported. Epsilon $1e-12$ prevents $\log(0)$.

64. What is IS blind to?

It never compares against real data, depends on this classifier, can miss within-class collapse, and can reward confident unrealistic or adversarial images.

65. Why was C3 IS approximately 1?

Every sample had nearly the same predicted class, so $p(y|x)$ and marginal $p(y)$ were nearly equal and the KL was near zero.

66. Derive FID.

Fit Gaussians to 128-D classifier features and compute $\| \mu_r - \mu_g \|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$.

67. What exact FID reference was used?

Every E row uses a seeded 5,000-image real training feature set as reference. The real row is a 5,000-image real test set against training, so it has FID 1.39865 rather than zero.

68. What is the FID floor experiment?

Two disjoint halves of the restricted real test set are compared, giving 1.200874. This estimates finite-sample feature-moment variation; it is not a universal mathematical lower bound.

69. Why can `sqrtn` return a complex result?

The covariance product is generally nonsymmetric and floating-point/singular calculations may introduce tiny imaginary components. The code checks magnitude, optionally retries with epsilon, rejects large imaginary parts, then takes the real part.

70. What happened numerically in this run?

Collapsed features could trigger a singular-covariance warning, but all distances remained finite; saved diagnostics show maximum imaginary component zero and no epsilon retry. A symmetric PSD method or diagonal regularization would be more robust.

71. Why is $\sigma=0$ FID not exactly zero?

Identical data/features should give zero theoretically. The saved 4.84e-09 is floating-point/square-root residue and is clamped nonnegative.

72. Why is the noise curve expected to be monotone here?

The assignment treats monotonicity as a required sanity check. With the same base images/noise and increasing sigma, a broadly increasing curve is expected and was observed from approximately 0 to 543.036. I would not claim universal mathematical monotonicity for every nonlinear learned feature map, clipping rule, and finite sample.

73. What is FID blind to?

It reduces chosen features to Gaussian first and second moments, has finite-sample bias, and can miss higher-order structure or artifacts not represented by the classifier features.

74. Why is C1 FID worse than fully collapsed C3?

C3's classifier-feature distribution happened to be closer to the real reference than C1's, despite having only one predicted mode. The grids suggest C3 is digit-like while C1 is high-frequency noise, but this is an interpretation consistent with the evidence, not a causal decomposition of the FID difference.

75. State the complete E table.

Real 5.7720+/-0.0262 / 1.399; C1 1.0347+/-0.0011 / 941.881; C3 1.0006+/-0.000025 / 821.241; D1 4.4772+/-0.1080 / 39.700; D2 4.6190+/-0.0898 / 17.117, where each pair is IS/FID.

H. Ownership, limitations, and debugging

76. What is the correct troubleshooting order?

Read saved curves, inspect shapes/ranges, overfit ten examples, run a short smoke test, and only then change architecture. The saved preflight logs show this process.

77. What are the main limitations?

Deterministic seed offsets rather than literal seed reuse; classifier proxies on generated/OOD data; no quantitative "samples stopped changing" criterion for C3; entangled A2 endpoints; singular-covariance risk; and no causal ablation isolating gradient penalty.

78. What assistance was used?

AI assistance helped inspect the handout/starter, draft standard PyTorch code, operate the Colab run, and assemble materials. It is disclosed. The measured claims come from the saved run, and ownership must be demonstrated by deriving, tracing, and modifying the code live.

79. What code should you be able to write unaided?

`kl_term`, reparameterization, the two generator losses, `mode_stats`, `gradient_penalty`, `inception_score`, and the FID mean/covariance calculation and checks.

80. What is the strongest correctness evidence?

No single result. Together: preflight overfits/smoke tests, full per-iteration logs, exact threshold row, fixed-noise comparisons, classifier confusion matrix, saved sample histograms, metric split CSVs, FID sanity checks, and agreement between report, JSON, CSVs, and figures.

Mock viva

Use a timer. Give the direct answer first, then equation/evidence, then one limitation. Score each answer 0-2: 0 incorrect, 1 conceptually right but unsupported, 2 correct and tied to this run.

Round 1: core concepts, 12 minutes

1. In 30 seconds, describe the complete assignment.
2. Derive the VAE KL expression from μ and $\log \text{var}$.
3. Why is the KL divided by 12,288 in your code?
4. Why does the AE decode its own codes but fail on normal draws?
5. Derive both logistic generator gradients with respect to the fake logit.
6. Why do the objectives share an equilibrium but behave differently early in training?
7. Derive reverse KL and the $\ln 6$ result.
8. State the WGAN-GP critic and generator objectives.
9. Derive IS and express it using entropies.
10. Explain every term in FID and two things it can miss.

Target: at least 18/20 with no missing equation.

Round 2: evidence challenge, 12 minutes

1. Prove your roll-derived values from $R=27$.
2. Open the exact C1 row used as the reference.
3. Calculate its 1% threshold and prove row 448 is the first crossing.
4. What happened at row 447, and why does it matter to your wording?
5. Show whether the reported threshold used the first-layer or full gradient.
6. Recompute C3's number of modes and reverse KL from raw counts.
7. Point to D1 and D2 proportions and describe their remaining imbalance.
8. Show one saved IS split and identify its tensor shapes.
9. Distinguish the real-training FID reference, the real row, and the floor.
10. Point to the preflight trace that best demonstrates natural debugging.

Target: all files opened in under 30 seconds each and at least 18/20.

Round 3: code ownership, 15 minutes

Write or explain without looking:

1. VAE reparameterization and KL.
2. Stable saturating and non-saturating loss lines.
3. `mode_stats`, including zero KL terms and the 1% rule.
4. WGAN-GP interpolation and penalty.
5. IS for one split.
6. FID means, covariances, $\sqrt{m}$, imaginary check, and epsilon retry.
7. Why `fake.detach()` and parameter freezing occur in different phases.
8. Why `create_graph=True` is necessary.

Target: no sign error and no confusion between probability/logit/critic score.

Round 4: hostile examiner follow-ups

"Your validation VAE loss is lower than training. Is that suspicious?"

Training reconstructs sampled z , while validation decodes μ ; BatchNorm modes also differ. The train/validation reconstruction values therefore are not under identical stochastic conditions. The fair AE/VAE comparison uses the validation reconstruction values.

"You violated the seed rule."

The base seed is 1327, and every offset is recorded. Offsets create deterministic independent substreams, but the approach is not literal compliance with resetting 1327 for every run. This is a reproducibility limitation, not something to hide.

"Your FID code produced a singular warning. Why trust it?"

Total collapse makes covariance low-rank. The result remained finite, the imaginary component was zero, and two sanity checks behaved correctly. Still, adding diagonal regularization proactively or using a symmetric PSD formulation would be a stronger implementation.

"WGAN-GP improved FID. Prove the gradient penalty caused it."

I cannot isolate that cause: objectives, critic steps, learning rate/betas, and critic BatchNorm also changed. The defensible conclusion is that the complete WGAN-GP configuration performed better in this fixed run.

"Why should I trust a classifier judging GAN samples?"

It achieved 99.019% on in-domain held-out digits and its confusion matrix is saved, but artifacts are OOD and can receive confident labels. Therefore I combine classifier statistics with sample grids and FID; none is sufficient alone.

"Did your C3 samples objectively stop changing?"

No quantitative temporal stopping metric was logged. The run ended after 3,000 outer iterations and showed complete final collapse by grid and 5,000-sample histogram. I should not claim more than that evidence.

"Why is real FID not zero?"

The real row is a real test subset compared with a separate real training reference. The floor is another comparison between two disjoint test halves. Only the $\sigma=0$ sanity check compares an image set with itself.

"Why are A4 and B2 missing?"

They are referenced but not defined in the supplied handout. I completed every explicit technical task and did not invent requirements.

Readiness standard

You are ready when you can complete all rounds twice on different days, score at least 90%, make no loss-sign or set-identity error, and open each evidence file without searching the entire folder.

Last-minute flashcards

Prompt	Answer			
Base facts?	R27, seed1327, held-out7, digits 0/2/3/6/7/9, Blond_Hair			
Main run?	20L-11327_main_v1			
AE/VAE setup?	latent64, Adam1e-3, batch64, 10 epochs, 27k/3k			
AE/VAE val MSE?	.006605 / .008301; gap .001696			
Reparameterization?	$z = \mu + \exp(.5 \log \text{var}) * \epsilon$			
KL?	.5 $\sum(\exp(\log \text{var}) + \mu^2 - 1 - \log \text{var})$			
KL scaling?	divide by $3 * 64 * 64$; beta remains 1			
PSNR?	$10 \log_{10}(1/\text{MSE})$ for range 1			
PSNR result?	13.3204 -> 19.4447 dB			
Classifier accuracy?	99.019%			
Saturating derivative?	-D(fake) wrt fake logit			
Non-saturating derivative?	-(1-D(fake))			
C1 reference?	iter100, 7.0218186e-05			
C1 threshold/death?	7.0218186e-07; iter448, zero			
C2 at 448?	first 3.83656, full 11.29571			
Gradient columns?	first G.fc parameter / all G parameters			
C3 recipe?	z2, no G BN, 5G:1D, lrG.002, lrD.0002			
C3 result?	all digit3, 1 mode, ln6=1.791759			
Coverage threshold?	>=1% = >=50 of 5000			
D1?	6 modes, RKL .052779, IS 4.4772, FID 39.700			
D2?	6 modes, RKL .019803, IS 4.6190, FID 17.117			
WGAN critic loss?	$E C(\text{fake}) - E C(\text{real}) + 10 \text{ GP}$			
WGAN G loss?	-E C(fake)			
GP?	$\text{mean}((\text{norm grad_xhat } C - 1)^2)$			
Why no critic BN?	batch coupling conflicts with per-sample gradient constraint			
IS?	$\exp(E \text{ KL}(p(y$ x) p(y)))			
Ideal six-class IS?	6; collapsed/identical = 1			
FID features?	classifier penultimate, 128-D			
FID reference?	5000 real training images			
Real IS/FID?	5.7720 / 1.39865			
C1 IS/FID?	1.0347 / 941.881			
C3 IS/FID?	1.0006 / 821.241			
FID floor?	1.200874, disjoint real test halves			
Noise FID?	0,14.144,100.552,326.750,543.036			
sqrtn complex?	nonsymmetric product + floating-point/singularity			
Highest-risk honesty point?	offsets are reproducible but not literal seed1327 every run			
Missing tasks?	A4, B2, Part F are referenced but undefined			
Debug order?	curves -> shapes -> overfit ten -> architecture			